Data Center RDIMM DDR5 Tester

HyperRAM
File: hyperram.kicad_sch

DDR5
File: DDR5.kicad_sch

Interfaces
File: interfaces.kicad_sch

Ethernet
File: ethernet.kicad_sch

Supply
File: supply.kicad_sch

Config SPI flash
File: config-spi.kicad_sch

FPGA power
File: fpga-power.kicad_sch

FPGA banks 12-15
File: fpga-banks-12-15.kicad_sch

FPGA banks 16-34
File: fpga-banks-16-34.kicad_sch

Logo N2 oshw_logo

Logo N1 antmicro_logo

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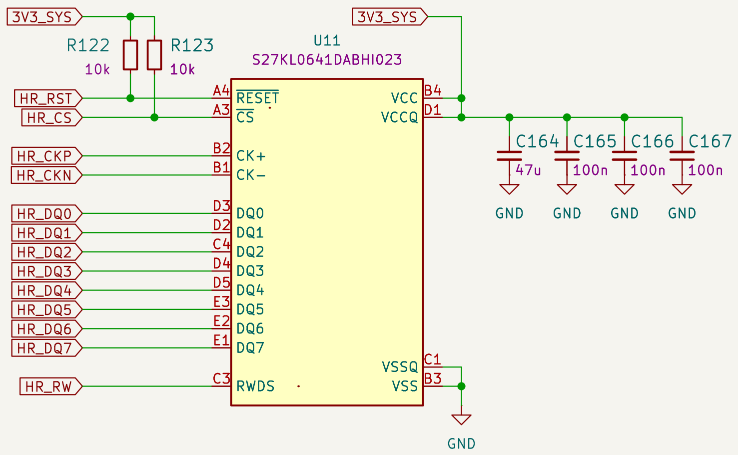
Sheet: /
File: data-center-rdimm-ddr5-tester.kicad_sch

Title: Data Center RDIMM DDR5 Tester

Size: A3 Date: 2023-07-27 **Rev: 1.0.1:85273**

KiCad E.D.A. 8.0.8 Id: 1/10

HyperRAM



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Size: A3 | Date: 2023-07-27 | **Rev: 1.0.1:85273**
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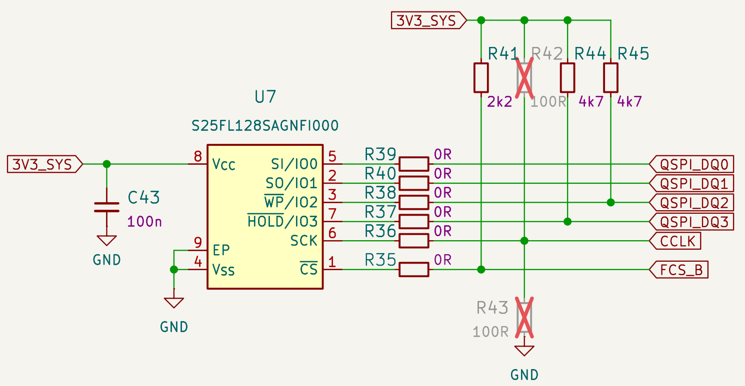
Master SPI Quad (x4) configuration scheme

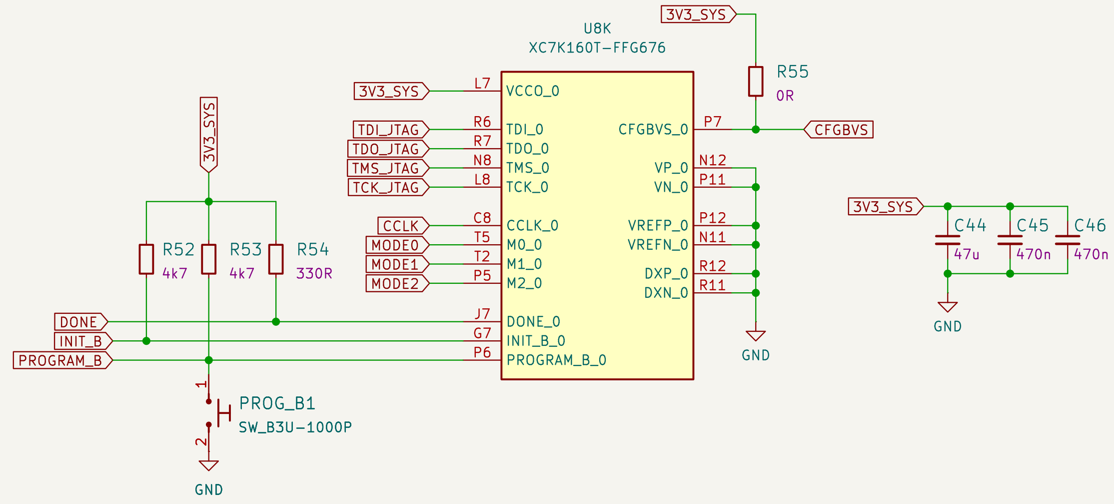
Follows Figure 2-14 7 Series FPGAs Configuration User Guide
UG470 (v1.13.1)

(Q)SPI flash



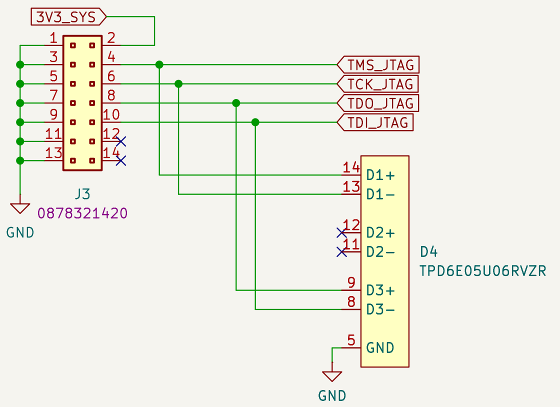
FPGA BANK 0

TODO: verify after FPGA swap

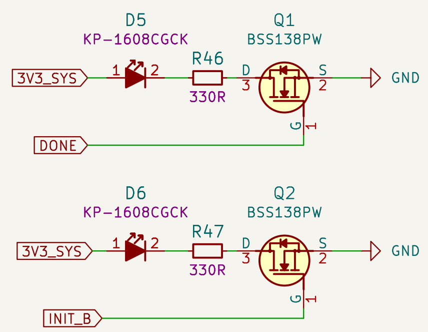


JTAG Connector

Compatible with Xilinx Platform Cable

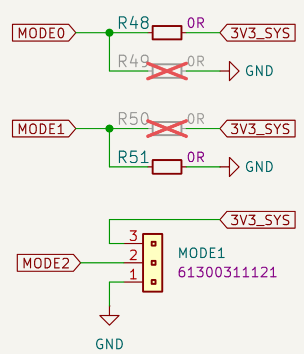


STATUS LEDs



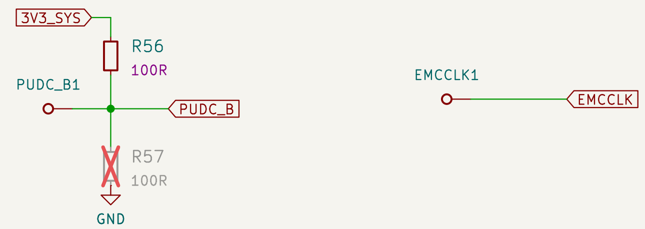
Configuration Modes

For details, see UG470 p. 21



Config. mode	MODE[2:0]
Master Serial	000
Master SPI	001
Master BPI	010
Master SelectMAP	100
JTAG	101
Slave SelectMAP	110
Slave Serial	111

Probes



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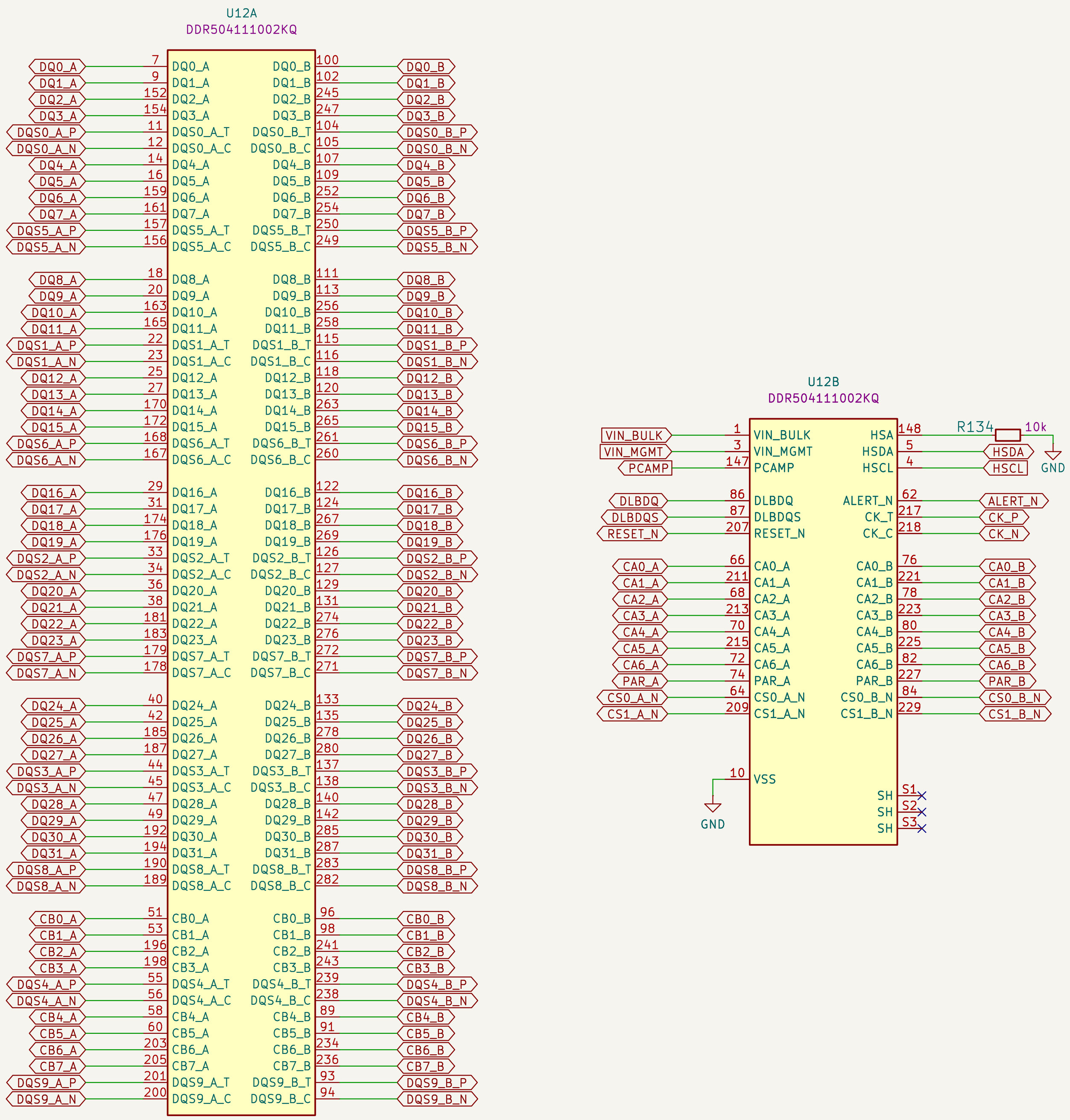
DDR5 RDIMM connector



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Size: A3	Date: 2023-07-27	Rev: 1.0.1:85273
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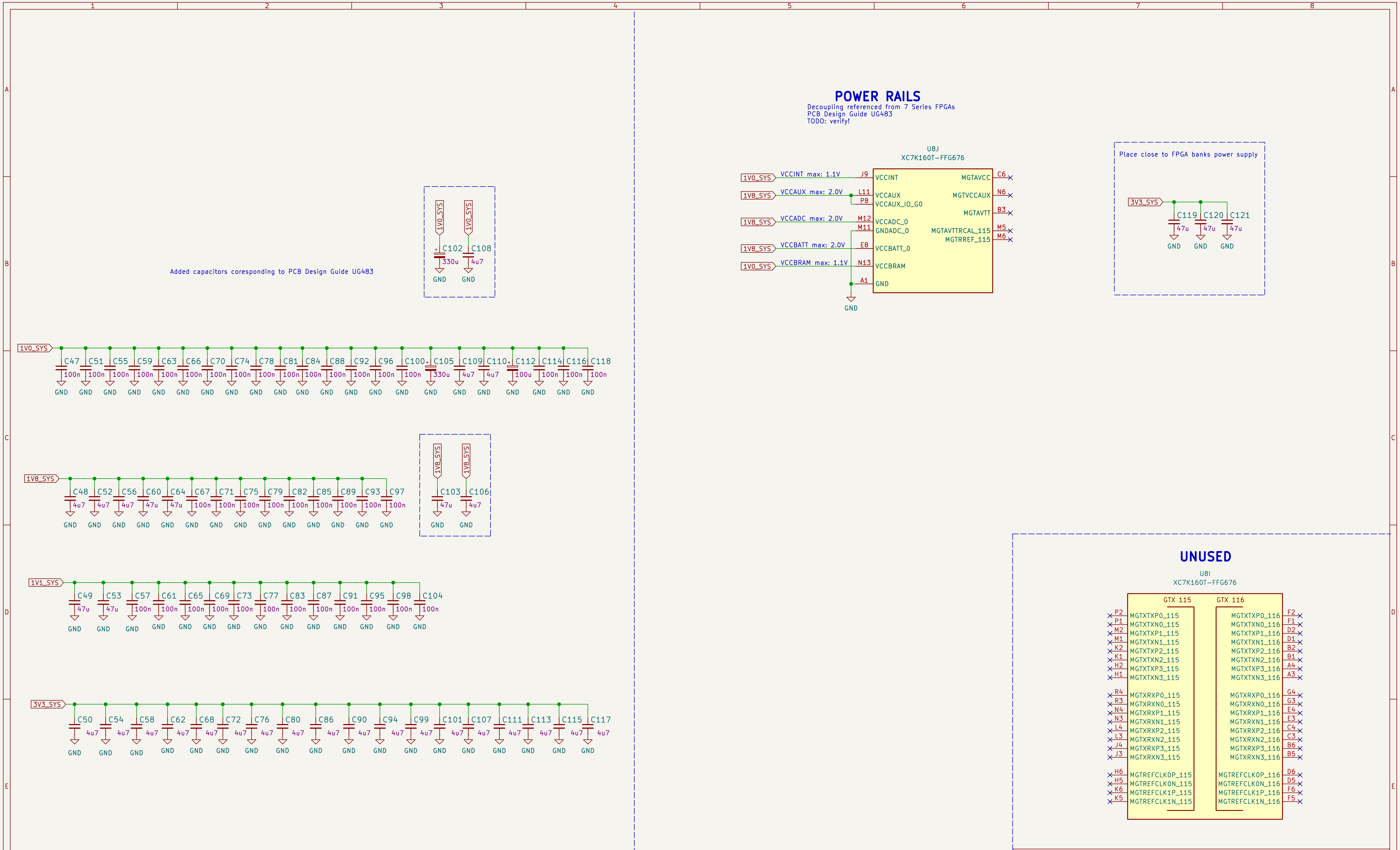

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Title: Data Center RDIMM DDR5 Tester

Size: A3	Date: 2023-07-27	**Rev: 1.0.1:85273**
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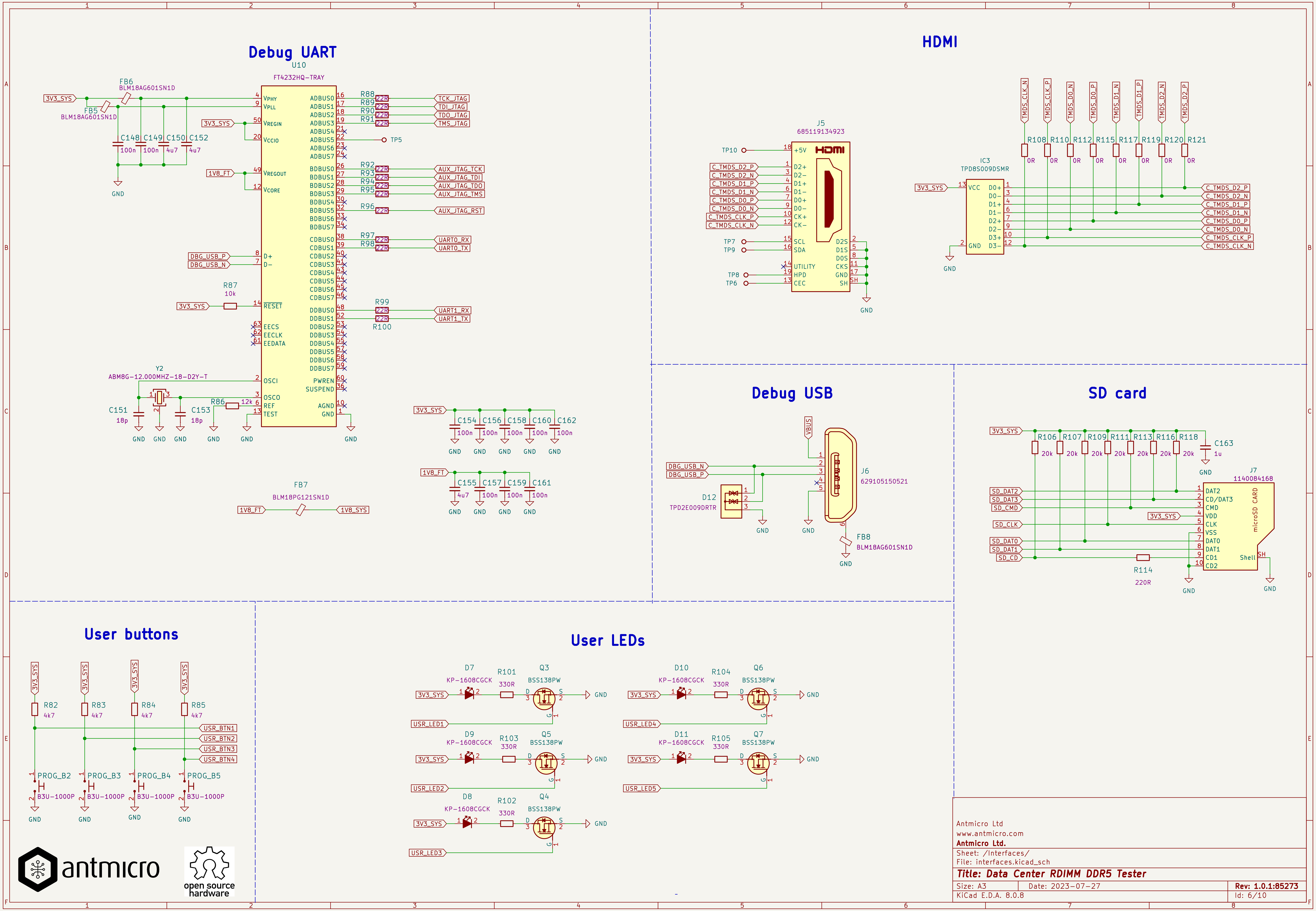

Debug UART
U10
FT4232HQ-TRAY
FB6
BLM18AG601SN1D
FB5
BLM18AG601SN1D
3V3_SYS
1V8_FT
C148 100n
C149 100n
C150 4u7
C152 4u7
GND
VPHY
VPLL
VREGIN
VCCIO
VREGOUT
VCORE
D+
D-
RESET
EECS
EECLK
EEDATA
OSCI
OSCO
REF
TEST
ADBUS0
ADBUS1
ADBUS2
ADBUS3
ADBUS4
ADBUS5
ADBUS6
ADBUS7
BDBUS0
BDBUS1
BDBUS2
BDBUS3
BDBUS4
BDBUS5
BDBUS6
BDBUS7
CDBUS0
CDBUS1
CDBUS2
CDBUS3
CDBUS4
CDBUS5
CDBUS6
CDBUS7
DDBUS0
DDBUS1
DDBUS2
DDBUS3
DDBUS4
DDBUS5
DDBUS6
DDBUS7
PWREN
SUSPEND
AGND
GND
R88
R89
R90
R91
22R
TCK_JTAG
TDI_JTAG
TDO_JTAG
TMS_JTAG
TP5
R92
R93
R94
R95
R96
AUX_JTAG_TCK
AUX_JTAG_TDI
AUX_JTAG_TDO
AUX_JTAG_TMS
AUX_JTAG_RST
R97
R98
UART0_RX
UART0_TX
R99
R100
UART1_RX
UART1_TX
DBG_USB_P
DBG_USB_N
R87
10k
Y2
ABM8G-12.000MHZ-18-D2Y-T
C151 18p
C153 18p
R86 12k
C154 100n
C156 100n
C158 100n
C160 100n
C162
C155 4u7
C157 100n
C159 100n
C161 100n
FB7
BLM18PG121SN1D
1V8_SYS
HDMI
J5
685119134923
TP10
+5V
C_TMDS_D2_P
C_TMDS_D2_N
C_TMDS_D1_P
C_TMDS_D1_N
C_TMDS_D0_P
C_TMDS_D0_N
C_TMDS_CLK_P
C_TMDS_CLK_N
D2+
D2-
D1+
D1-
D0+
D0-
CK+
CK-
TP7
TP9
SCL
SDA
UTILITY
HPD
CEC
TP8
TP6
D2S
D1S
D0S
CKS
SH
TMDS_CLK_N
TMDS_CLK_P
TMDS_D0_N
TMDS_D0_P
TMDS_D1_N
TMDS_D1_P
TMDS_D2_N
TMDS_D2_P
R108
R110
R112
R115
R117
R119
R120
R121
0R
IC3
TPD8S009DSMR
VCC
D3+
D3-
Debug USB
VBUS
DBG_USB_N
DBG_USB_P
D12
TPD2E009DRTR
J6
629105150521
FB8
BLM18AG601SN1D
SD card
R106
R107
R109
R111
R113
R116
R118
20k
C163
1u
J7
1140084168
microSD CARD
SD_DAT2
SD_DAT3
SD_CMD
SD_CLK
SD_DAT0
SD_DAT1
SD_CD
DAT2
CD/DAT3
CMD
VDD
CLK
VSS
DAT0
DAT1
CD1
CD2
Shell
R114
220R
User buttons
R82
R83
R84
R85
4k7
USR_BTN1
USR_BTN2
USR_BTN3
USR_BTN4
PROG_B2
PROG_B3
PROG_B4
PROG_B5
B3U-1000P
User LEDs
D7
D8
D9
D10
D11
KP-1608CGCK
R101
R102
R103
R104
R105
330R
Q3
Q4
Q5
Q6
Q7
BSS138PW
USR_LED1
USR_LED2
USR_LED3
USR_LED4
USR_LED5
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Sheet: /Interfaces/
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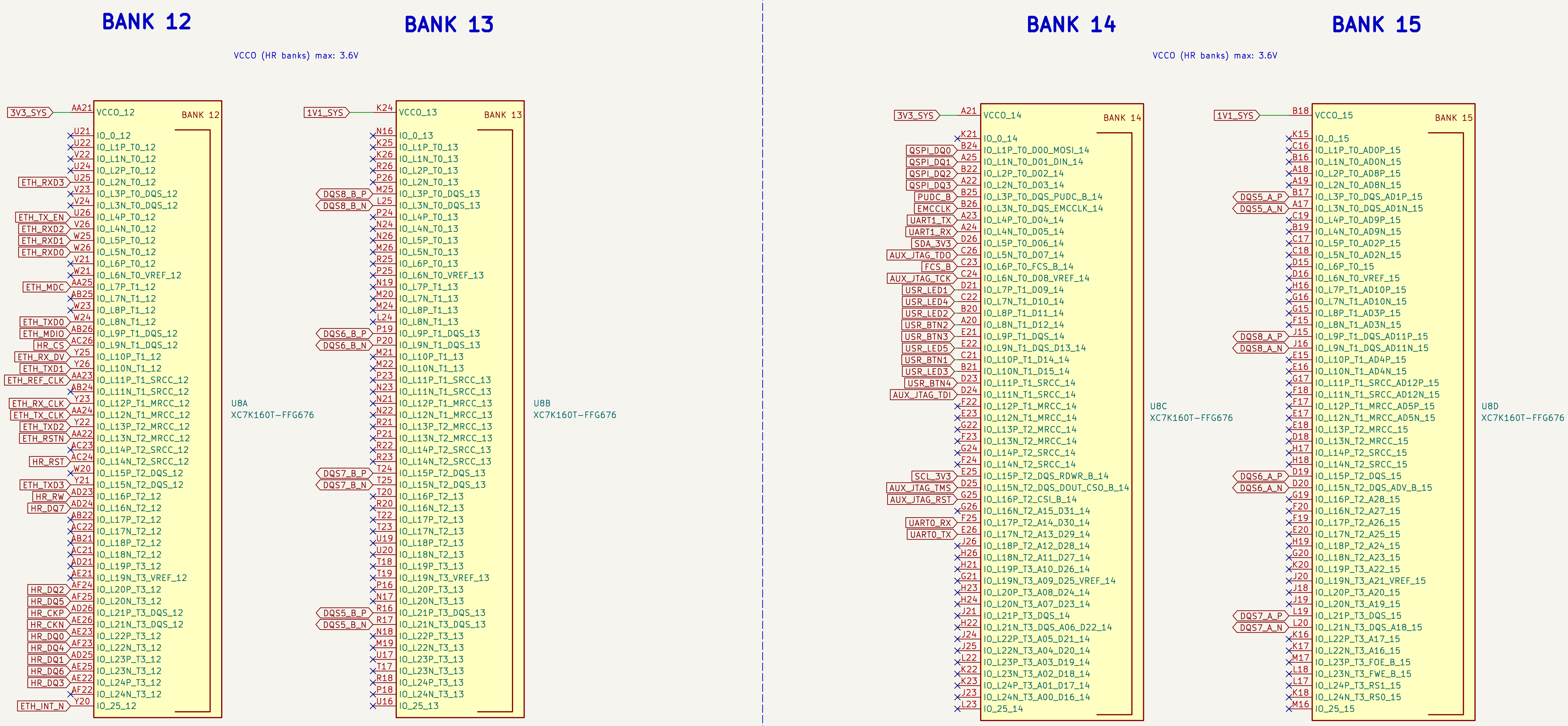
Size: A3
Date: 2023-07-27
Rev: 1.0.1:85273
KiCad E.D.A. 8.0.8
Id: 6/10



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File: fpga-banks-12-15.kicad_sch

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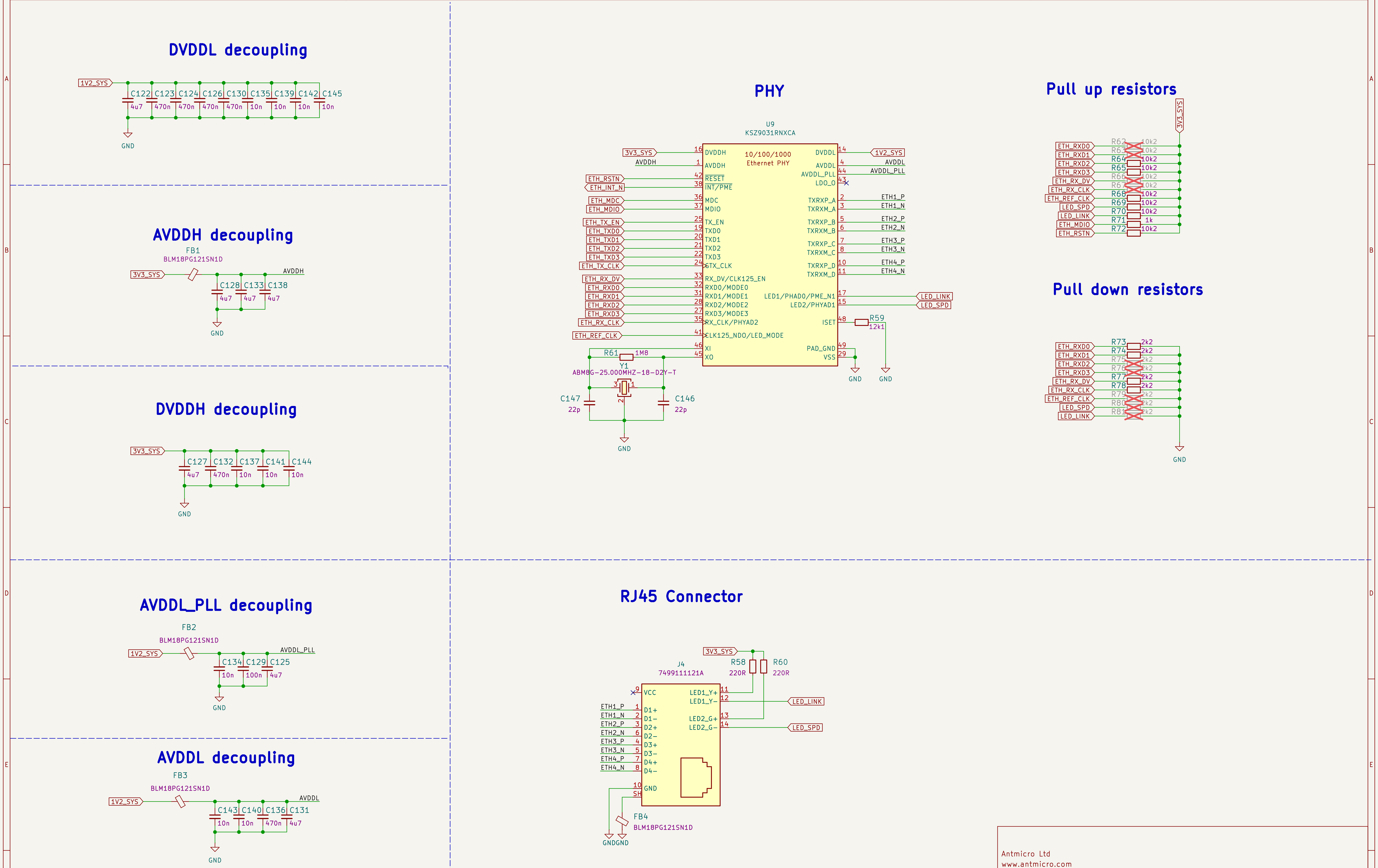
Size: A3	Date: 2023-07-27	**Rev: 1.0.1:85273**
KiCad E.D.A. 8.0.8		Id: 7/10



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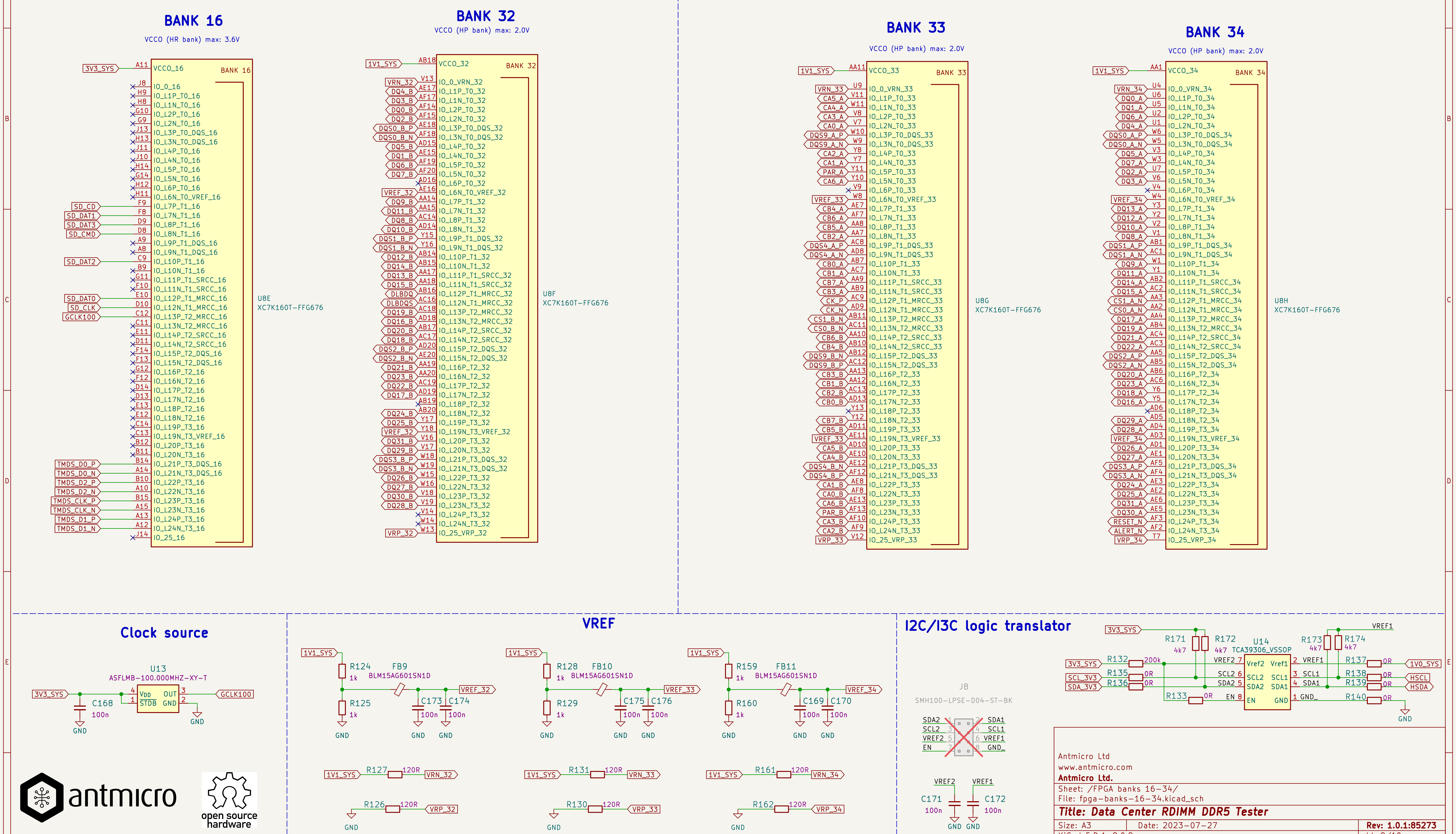

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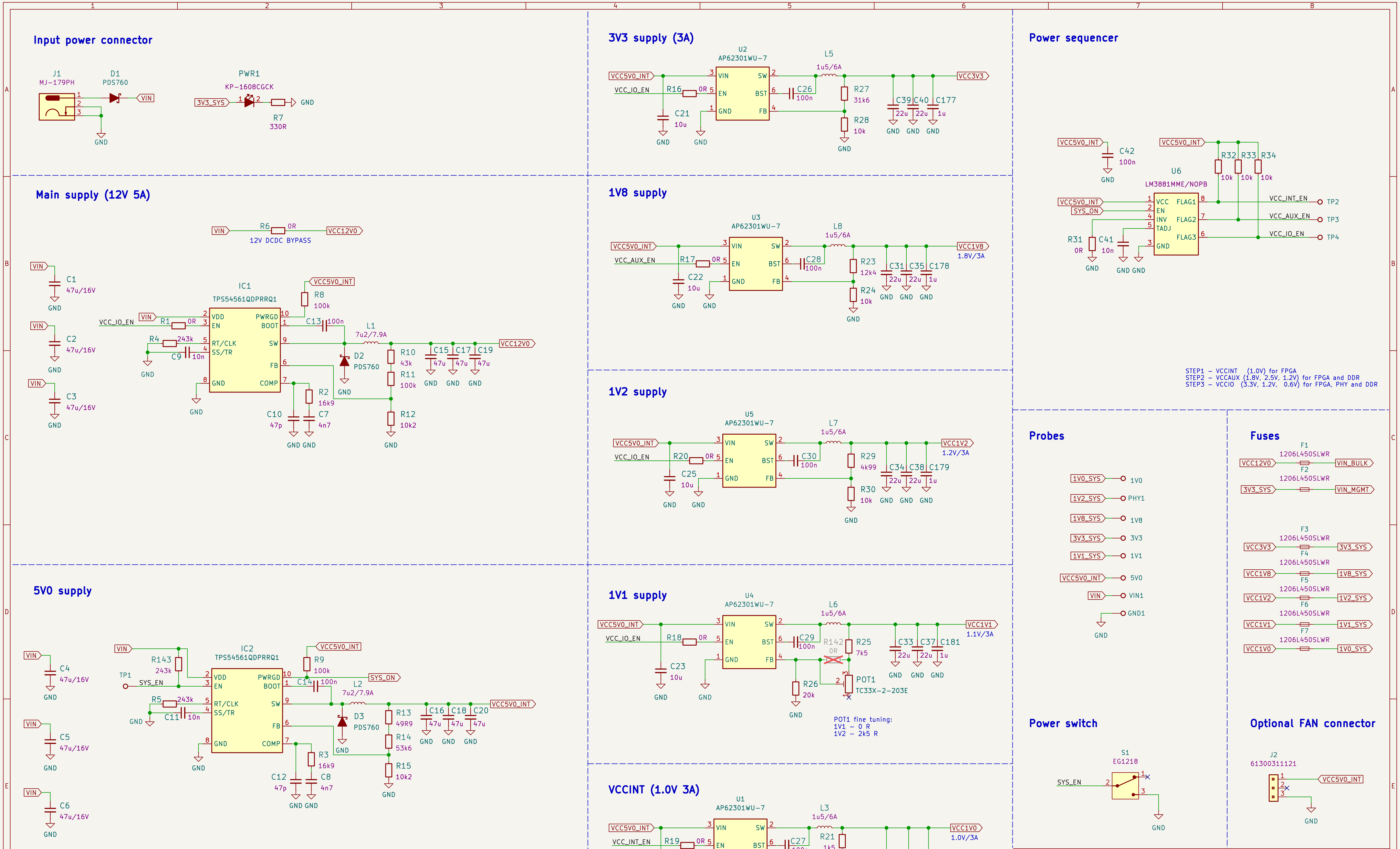

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Sheet: /Supply/
File: supply.kicad_sch

Title: Data Center RDIMM DDR5 Tester

Size: A3 | Date: 2023-07-27 | Rev: 1.0.1:85273
KiCad E.D.A. 8.0.8 | Id: 10/10